

Brouwer-Lefschetz Example Figure 8 complex

G : finite abstract simplicial complex.

$T : G \rightarrow G$ is a simplicial map if
 $x \subset y \Rightarrow T(x) \subset T(y)$ and $T(V) \subset V$

$\chi_T(G) = \text{str}(U_T | \ker(H))$ **Lefschetz number**

$Uf(x) = f(T(x))\text{sign}(T|x)$ **Koopman**

$F = \{x \in G, T(x) = x\}$ **fixed points**

$i_T(x) = \text{sign}(T|x)\omega(x)$ **index**

Lefschetz fixed point Theorem

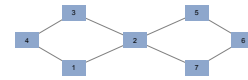
$$\chi_T(G) = \sum_{x \in F} i_T(x)$$

Brouwer fixed point Theorem

T on contractible G has fixed point

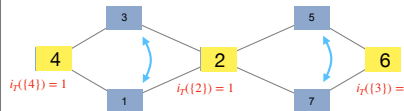
$G = \{[1], [2], [3], [4], [5], [6], [7], [1,2], [1,4], [2,3], [2,5], [2,7], [3,4], [5,6], [6,7]\}$

$\text{Aut}(G) = D_4$
dihedral group



cycles	permutation T	$\chi_T(G)$
()	= (1 2 3 4 5 6 7)	-1
(57)	= (1 2 3 4 7 6 5)	1
(13)	= (3 2 1 4 5 6 7)	1
(13)(57)	= (3 2 1 4 7 6 5)	3
(15)(37)(46)	= (5 2 7 6 1 4 3)	1
(1537)(46)	= (5 2 7 6 3 4 1)	1
(1735)(46)	= (7 2 5 6 1 4 3)	1
(17)(35)(46)	= (7 2 5 6 3 4 1)	1

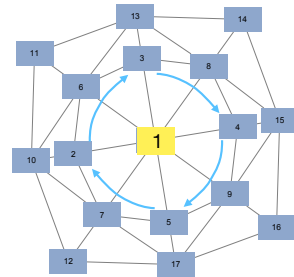
T	indices $i_T(x)$
(1,2,3,4,5,6,7)	{1,1,1,1,1,1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1}
(1,2,3,4,7,6,5)	{1,1,1,1,1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1}
(3,2,1,4,5,6,7)	{1,1,1,1,1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1}
(3,2,1,4,7,6,5)	{1,1,1}
(5,2,7,6,1,4,3)	{1}
(5,2,7,6,3,4,1)	{1}
(7,2,5,6,1,4,3)	{1}
(7,2,5,6,3,4,1)	{1}



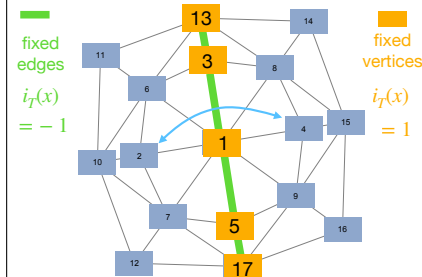
Example

Ball

$\text{Aut}(G) = D_4$ $\chi_T(G)=1$ for all T



$T = (2\ 3\ 4\ 5)(6\ 8\ 9\ 7)(10\ 13\ 15\ 17)(11\ 14\ 16\ 12)$



$T = (2\ 4)(6\ 8)(7\ 9)(10\ 15)(11\ 14)(12\ 16)$

Proof

T is invertible on

$K = \bigcap_{k \geq 0} T^k(G)$ **the attractor**

$Uf(x) = f(T(x))\text{sign}(T|x)$
unitary Koopman operator

Heat deformation argument:

$\text{str}(Ue^{-tH})$ is t independent
(McKean-Singer symmetry)

$t=0$ $\text{str}(U) = \sum_{x \in F} i_T(x)$

add up $\omega(x)\text{sign}(T|x)$ over diagonal of U , if x is a fixed point.

$t \rightarrow \infty$ $\text{str}(Ue^{-tH}) \rightarrow \chi_T(G)$

H and U commute.

QED

Quotient

G : finite abstract simplicial complex.

$A \subset \text{Aut}(G)$ subgroup

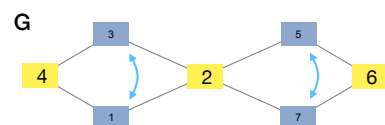
$$\frac{1}{|A|} \sum_{T \in A} \chi_T(G) \quad \text{average Lefschetz number}$$

$$\chi(G/A) = \frac{1}{|A|} \sum_{T \in A} \chi_T(G)$$

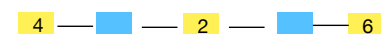
Example: $A = \{1, T\}$ with $T = (13)(57)$

$$\chi_T(G) = 3 \quad \chi_1(G) = -1$$

average $(3-1)/2 = 1$



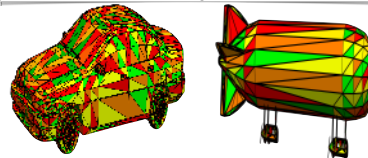
G/A path complex $\chi(G/A) = 1$



G is a branched cover over G/A !

On spheres:

An orientation preserving map of an object with the topology of a 2-sphere has at least 2 fixed simplices.



To the continuum

The discrete implies the continuum

but more is needed:

G has manifold structure
 T smooth
 isolated fixed points

Every closed set can be the fixed point set of a smooth map

The Mandelbrot set for example is the fixed point set of a smooth map

